Basics 5: Partial Differential Equations

I - Mathematical and computational foundations

Classification, data requirements, characteristics, conservation form, and well-posedness

Textbook draft, version 1.0. Most plasma simulations solve partial differential equations. Their mathematical type predicts information propagation, boundary data, regularity, and suitable numerical methods. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Identify independent variables, dependent fields, order, and linearity.
- Classify canonical elliptic, parabolic, and hyperbolic equations.
- Choose appropriate initial and boundary conditions.
- Derive characteristics for advection.
- Distinguish conservative and nonconservative forms.
- Explain weak solutions, eigenmodes, and well-posedness.

2 Prerequisites and reading strategy

- Basics 1-3 or equivalent multivariable calculus.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Definitions and examples

A PDE relates a field to its partial derivatives:

$$\mathcal{F}(\mathbf{x}, t, u, \partial_t u, \nabla u, \nabla^2 u, \ldots) = 0. \tag{3.1}$$

The order is the highest derivative. The equation is linear if u and its derivatives enter linearly with coefficients independent of u. Poisson, diffusion, wave, ideal-MHD, and kinetic equations illustrate different mathematical structures.

4 Elliptic, parabolic, and hyperbolic types

For

$$Au_{xx} + 2Bu_{xy} + Cu_{yy} + \cdots = 0, \tag{4.1}$$

the sign of $B^2 - AC$ gives the local classification. Poisson equation is elliptic and globally constrained by boundaries. Diffusion is parabolic and smooths gradients. The wave equation is hyperbolic and carries finite-speed signals. Ideal MHD is a nonlinear hyperbolic evolution system, while equilibrium calculations are closer to elliptic boundary-value problems.

5 Initial and boundary conditions

A first-order-in-time equation generally needs one initial field. A second-order spatial operator needs boundary conditions. Common forms are

$$u = g \qquad \text{Dirichlet,} \tag{5.1}$$

$$\frac{\partial u}{\partial n} = h \qquad \text{Neumann,} \tag{5.2}$$

$$au + b\frac{\partial u}{\partial n} = c \qquad \text{Robin.} \tag{5.3}$$

Data must be compatible with the operator. Pure-Neumann Poisson problems need an integral compatibility condition and have an arbitrary additive constant.

A problem is well posed if a solution exists, is unique, and depends continuously on data. Numerical sophistication cannot repair a fundamentally ill-posed model.

6 Characteristics and causality

For advection,

$$u_t + au_x = 0, \tag{6.1}$$

follow a curve $x(t)$:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = u_t + \dot{x}u_x. \tag{6.2}$$

Choosing $\dot{x} = a$ gives $du/dt = 0$, so

$$u(x,t) = u_0(x - at). \tag{6.3}$$

Characteristics identify causal directions and inflow boundaries. For $a > 0$, the left boundary is inflow and normally requires data; the right boundary is outflow.

7 Conservative form and discontinuities

A scalar conservation law is

$$u_t + \frac{\partial}{\partial x}F(u) = 0. \tag{7.1}$$

For smooth u, it is equivalent to $u_t + F'(u)u_x = 0$. Across a discontinuity, the conservative form determines shock speed:

$$s = \frac{F(u_R) - F(u_L)}{u_R - u_L}. \tag{7.2}$$

MHD supports shocks, so flux-form discretization is essential. Expanding products before discretization can destroy the correct jump condition.

8 Separation of variables, spectra, and weak forms

For the fixed-end wave equation, $u = X(x)T(t)$ gives

$$\frac{T''}{c^2 T} = \frac{X''}{X} = -k^2, \tag{8.1}$$

and boundary conditions select $k_n = n\pi/L$. This is the prototype of a normal-mode eigenproblem.

When solutions are not classically differentiable, multiply by a test function and integrate to obtain a weak form. Finite elements are based on weak forms. Standard PINNs often minimize strong residuals at collocation points, which may be inappropriate near shocks or ideal continuum singularities.

9 Computational laboratory

Implement one-dimensional Poisson, diffusion, and advection problems on a common grid. Use a direct boundary-value solve, explicit and implicit diffusion stepping, and upwind advection. Demonstrate global boundary influence for Poisson, a stability limit for explicit diffusion, and oscillations from centered advection. Measure convergence against analytic solutions.

10 Summary

- PDE type predicts information propagation and required data.
- Elliptic, parabolic, and hyperbolic equations behave qualitatively differently.
- Characteristics encode causality.
- Conservative form is essential for discontinuous solutions.
- Weak and strong residual formulations are not interchangeable in all regimes.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
$u(\mathbf{x}, t)$	field	Unknown dependent variable.
$\mathcal{F}$	PDE operator	Relation among field and derivatives.
$F(u)$	flux	Transported quantity in a conservation law.
s	shock speed	Discontinuity propagation speed.
$\partial_n u$	normal derivative	$\nabla u \cdot \mathbf{n}$.

11 Exercises

1. Classify $u_{xx} + 4u_{xy} + u_{yy} = 0$.
2. Derive characteristics for $u_t + a(x, t)u_x = 0$.
3. Find the compatibility condition for a pure-Neumann Poisson problem.
4. Derive Burgers shock speed.
5. Explain why a wave equation needs two initial conditions while advection needs one.

12 References

References

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